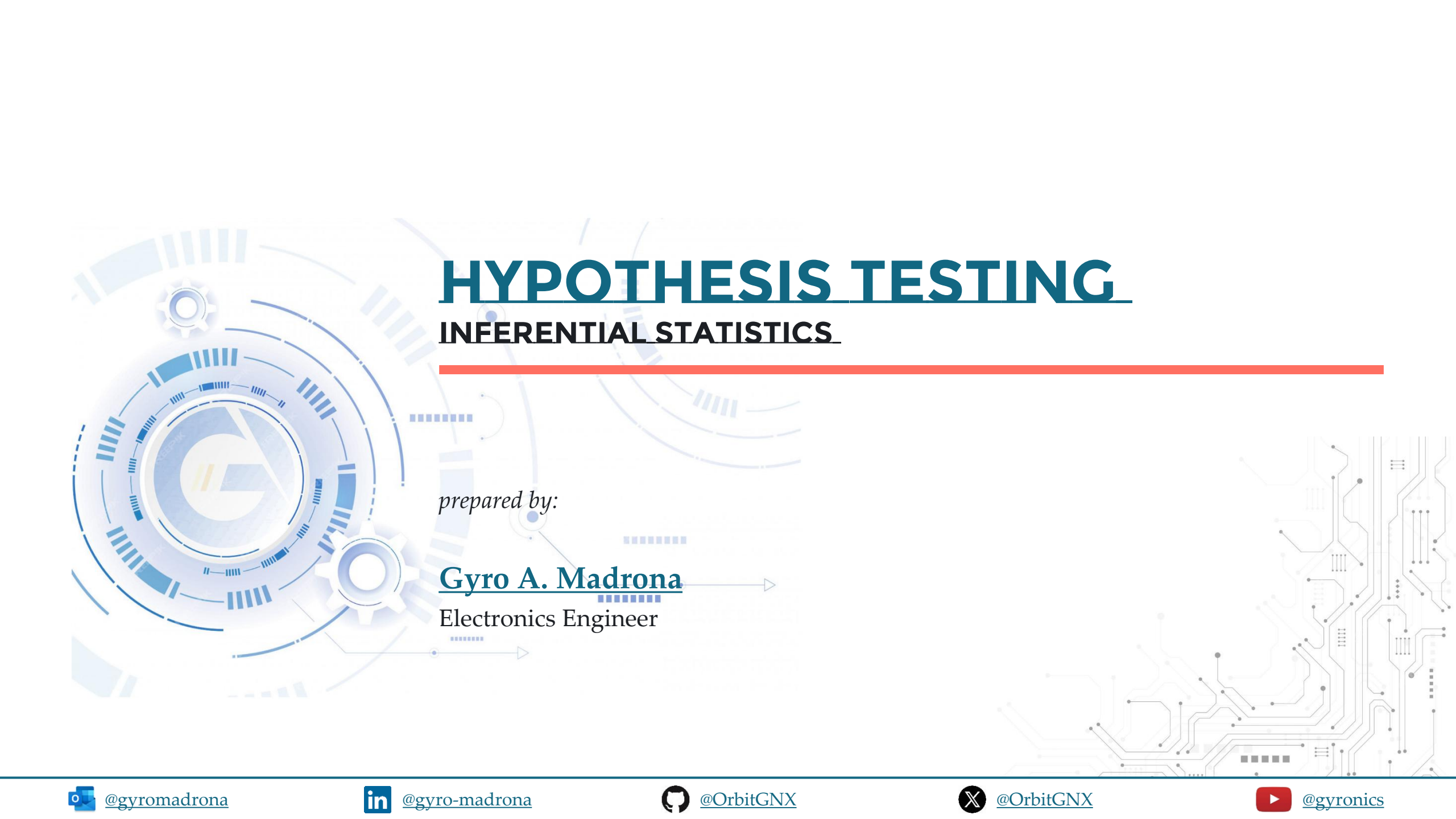




HYPOTHESIS TESTING

INFERENTIAL STATISTICS

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TOPIC OUTLINE

Hypothesis Test

Rejection Region

Critical Value and Z-score

p-Value



HYPOTHESIS TEST



HYPOTHESIS

A hypothesis is an initial assumption formed before collecting data, and it serves as a statement about a population parameter rather than about the sample data.



HYPOTHESIS TEST

A hypothesis test is simply comparing reality to an assumption and asking, “Did things change?”

Null Hypothesis (H_0)

Represents no change, no effect, or the status quo.

Alternative Hypothesis (H_a)

Represents the possibility that things did change or that there is a significant difference.



IS YOUR DATA GUILTY?

Hypothesis testing is like a legal system where the defendant is assumed innocent until proven guilty.

H_0 : Innocent



The Truth

	Innocent	Guilty
Accept	✓	✗
Reject	✗	✓

TYPES OF ERROR

1. Type I Error

The probability of rejecting the null hypothesis when it is true (α).

2. Type II Error

The probability of failing to reject the null hypothesis when it is false (β).



H_0 : No change

	The Truth	
	H_0 is true	H_0 is false
Fail to reject	✓	Type II Error (False negative)
Reject	Type I Error (False positive)	✓



POWER

The **power of a test** is the probability of correctly rejecting H_0 when it is false.

Formula

$$\text{power} = (1 - \beta)$$



The Truth

	The Truth	
	H_0 is true	H_0 is false
H_0 : No Change	Fail to reject	✓ Type II Error (β)
	Reject	Type I Error (α) ($1 - \beta$)



POWER

The power of a test is the probability of correctly rejecting H_0 when it is false.

Formula

$$\text{power} = (1 - \beta)$$

syntax

```
from statsmodels.stats.power
```

```
import TTestPower
```

standardized mean difference

Cohen's d = $(\bar{x} - \mu) / \sigma$

```
power = TTestPower().power(
```

```
    effect_size = cohen's d,
```

```
    nobs = sample size,
```

```
    alpha = significance level
```

```
)
```



EXERCISE

A company claims that the average lifespan of their batteries is 500 hours. A consumer group suspects that the average lifespan is **different** from **500 hours**.

Null Hypothesis

$$H_o: \mu_1 = 500$$

The average battery lifespan is 500 hours

Alternative Hypothesis

$$H_a: \mu_1 \neq 500$$

The average battery lifespan differs from 500 hours



EXERCISE

A company claims that the average lifespan of their batteries is 500 hours. A consumer group suspects that the batteries last **fewer than 500 hours**.

Null Hypothesis

$$H_o: \mu_1 \geq 500$$

The average battery lifespan is at least 500 hours

Alternative Hypothesis

$$H_a: \mu_1 < 500$$

The average battery lifespan is fewer than 500 hours



EXERCISE

A company claims that the average lifespan of their batteries is 500 hours. An independent lab believes that the batteries last **longer than 500 hours**.

Null Hypothesis

$$H_o: \mu_1 \leq 500$$

The average battery lifespan is 500 hours at most

Alternative Hypothesis

$$H_a: \mu_1 > 500$$

The average battery lifespan is longer than 500 hours



REJECTION REGION



SIGNIFICANCE LEVEL

The significance level (α) determines the threshold for deciding whether to reject the null hypothesis (H_0).

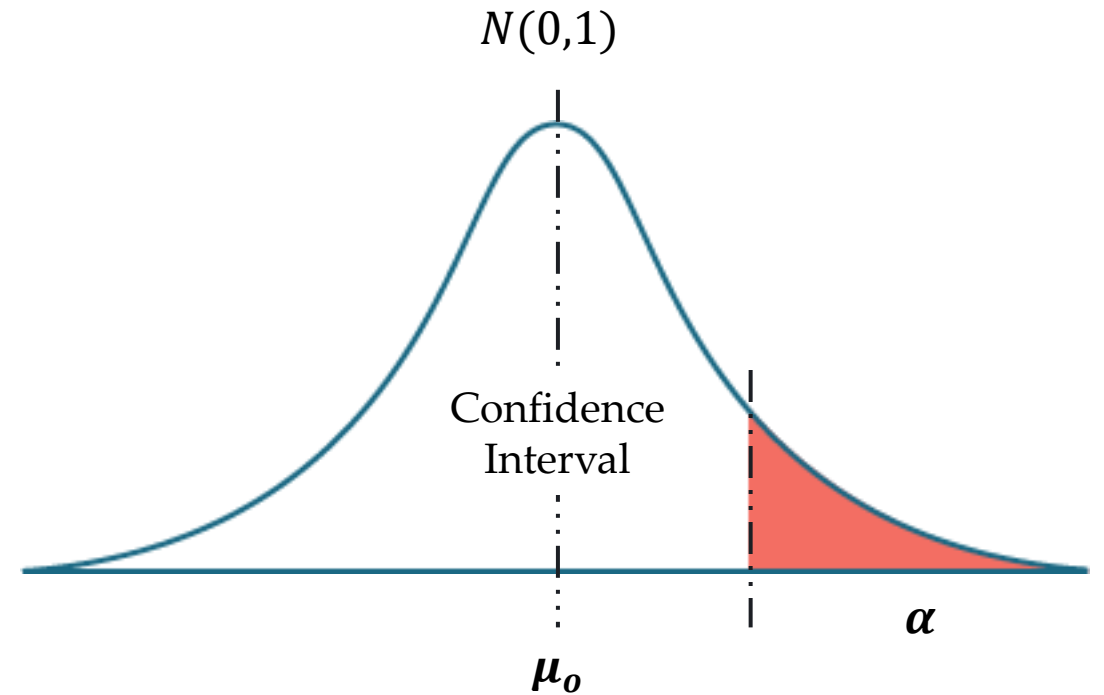
Typical values for α

0.01

0.05

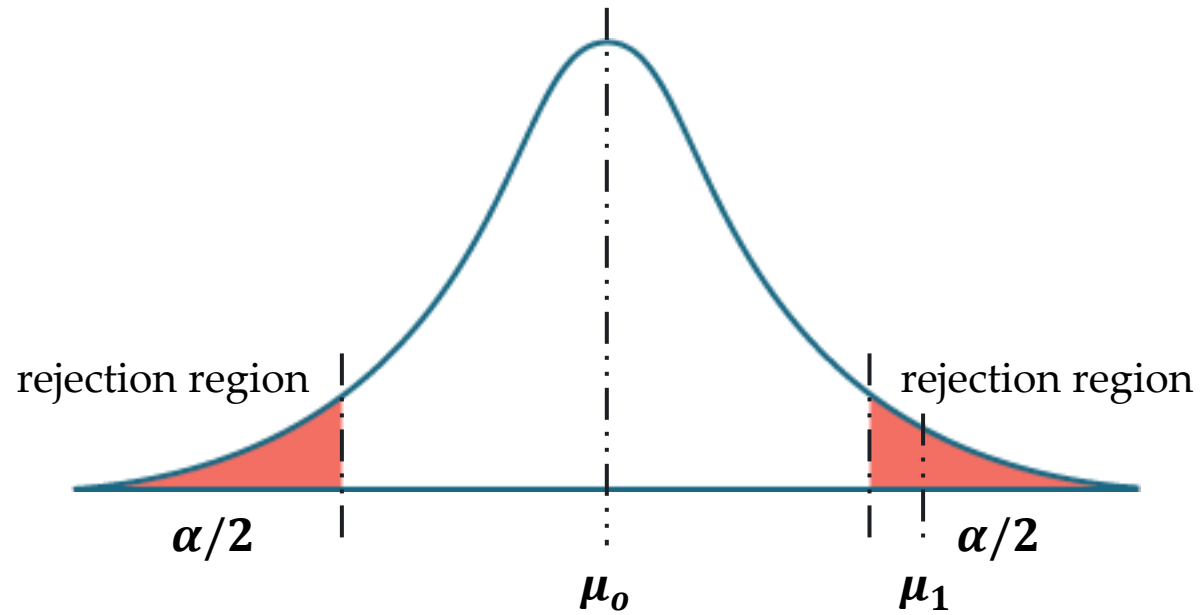
0.1

Standard Normal Distribution



REJECTION REGION

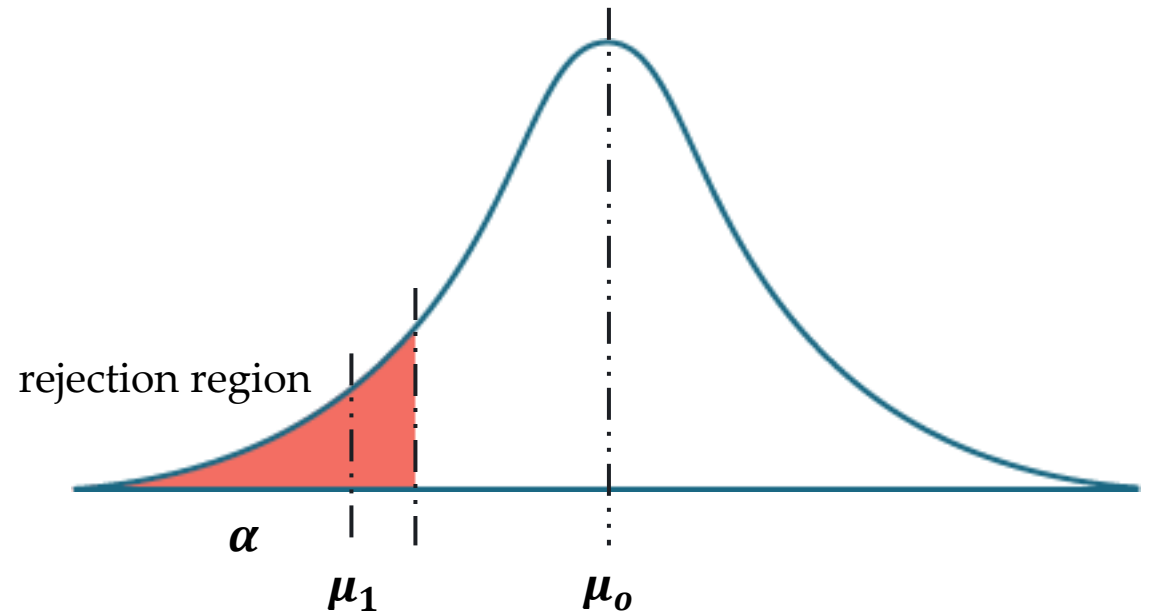
Two-Tailed Test



$$H_o: \mu_1 = \mu_o$$

$$H_a: \mu_1 \neq \mu_o$$

Left-Tailed Test



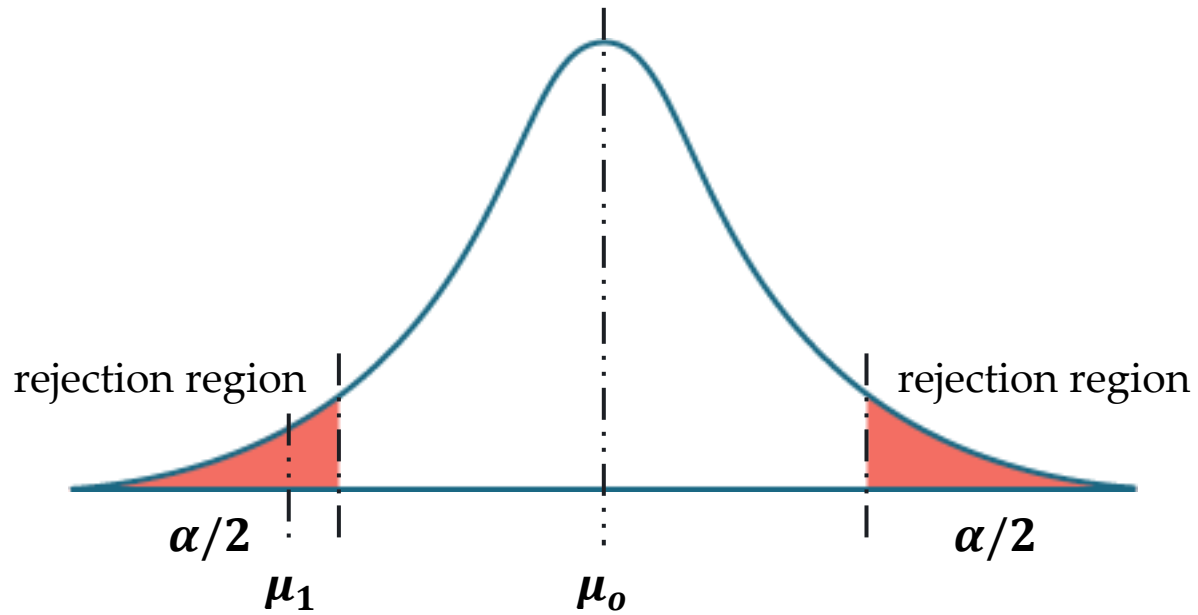
$$H_o: \mu_1 \geq \mu_o$$

$$H_a: \mu_1 < \mu_o$$



REJECTION REGION

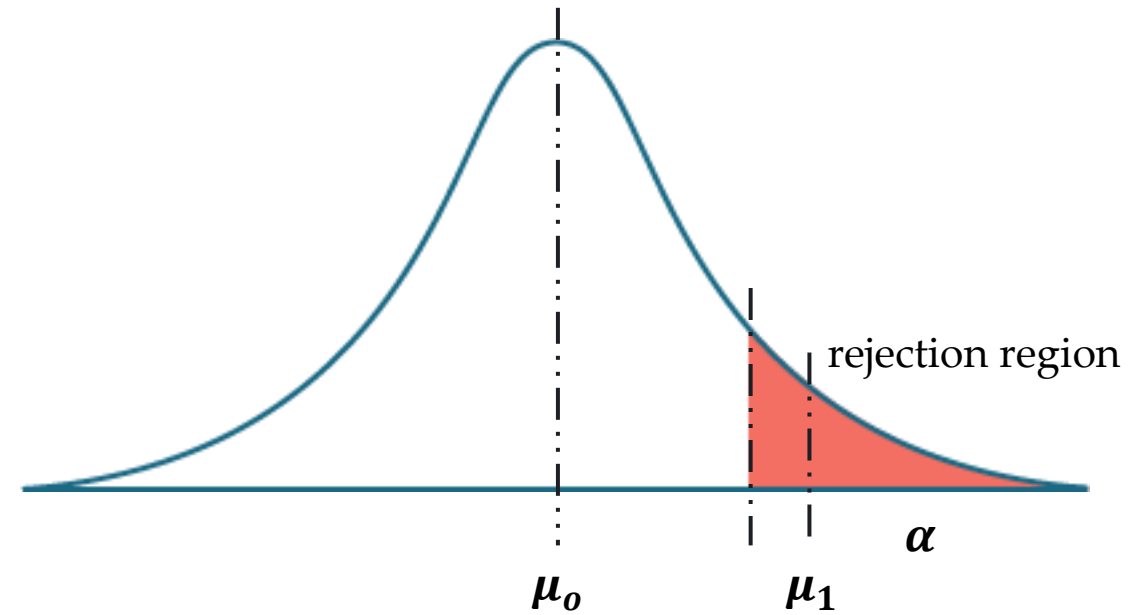
Two-Tailed Test



$$H_o: \mu_1 = \mu_o$$

$$H_a: \mu_1 \neq \mu_o$$

Right-Tailed Test



$$H_o: \mu_1 \leq \mu_o$$

$$H_a: \mu_1 > \mu_o$$



CRITICAL VALUE AND Z-SCORE



CRITICAL VALUE AND Z-SCORE

Right-Tailed Test

lowercase z

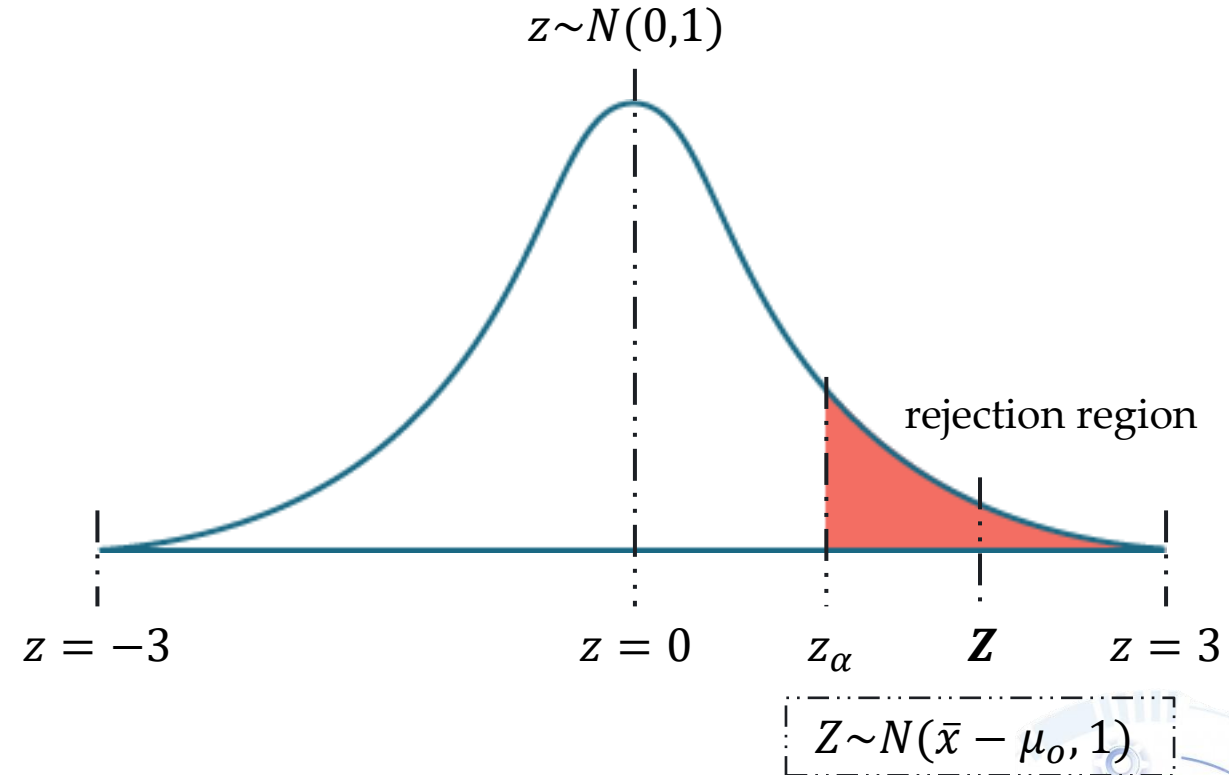
z refers to the **critical value** obtained from the standard normal distribution table (z-table).

uppercase Z

Z is a standardized variable associated with the test called the **z-score**.

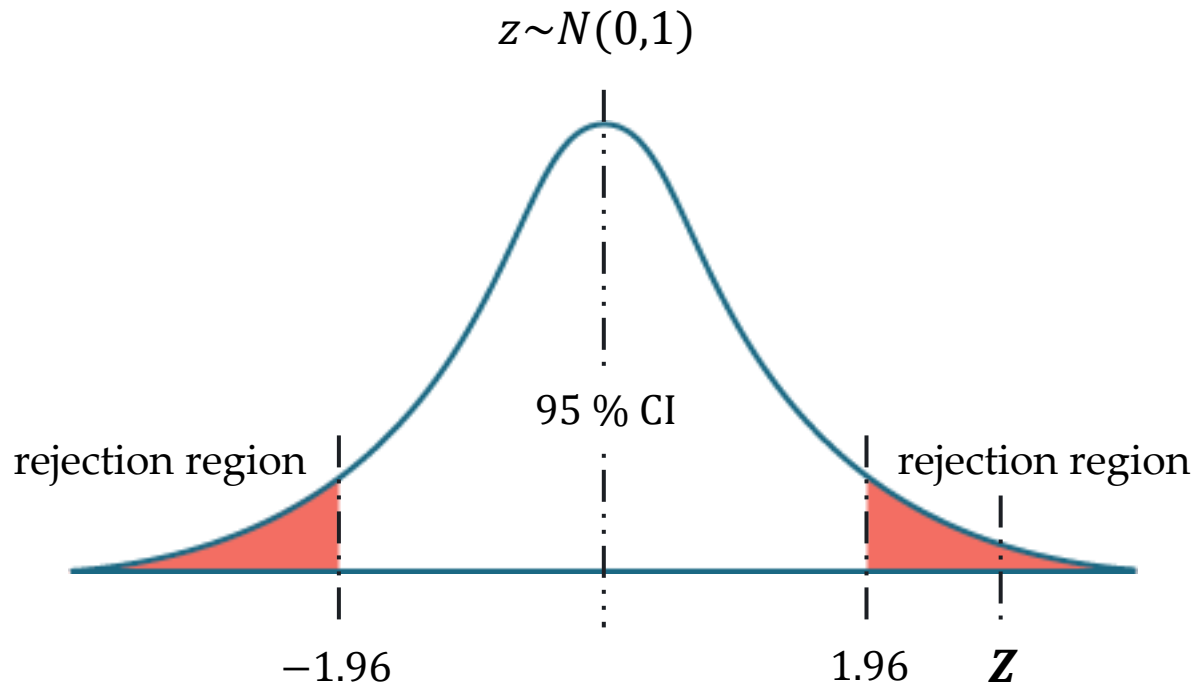
Formula

$$Z = \frac{\bar{x} - \mu_o}{\sigma/\sqrt{n}}$$



EXERCISE

Two-Tailed Test



$$\alpha = 0.05$$

$$z_{0.025} = 1.96$$

Null Hypothesis

$$H_0: \mu_1 = 500$$

The average battery lifespan is 500 hours

Alternative Hypothesis

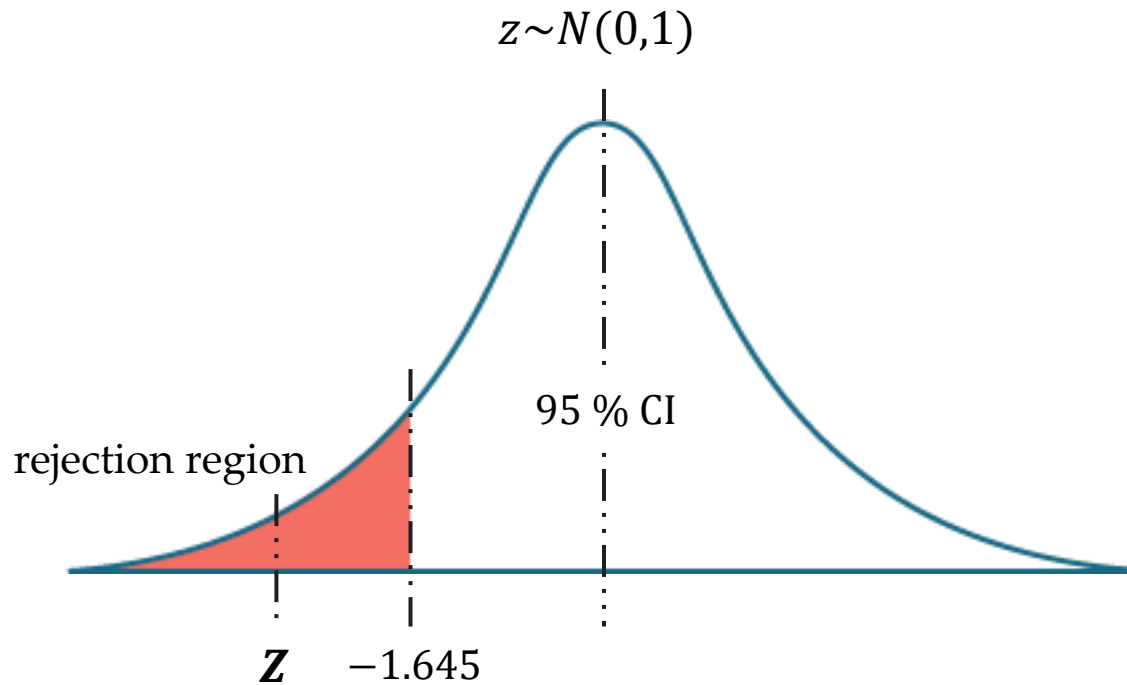
$$H_a: \mu_1 \neq 500$$

The average battery lifespan differs from 500 hours



EXERCISE

Left-Tailed Test



$$\alpha = 0.05$$

$$z_{0.05} = 1.645$$

Null Hypothesis

$$H_0: \mu_1 \geq 500$$

The average battery lifespan is at least 500 hours

Alternative Hypothesis

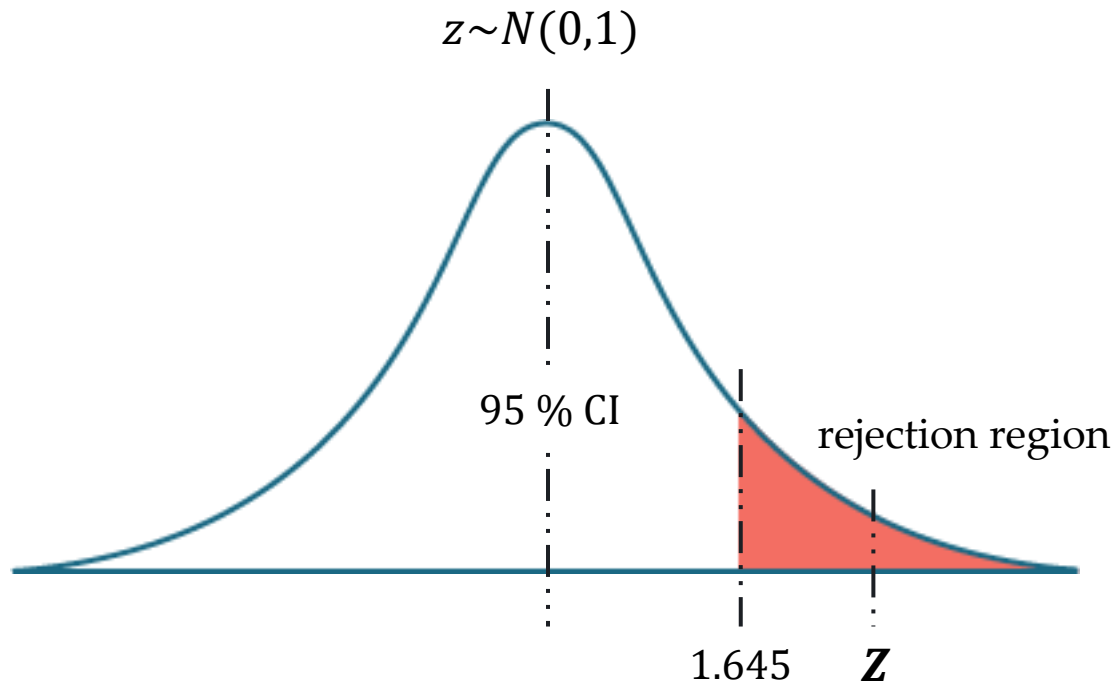
$$H_a: \mu_1 < 500$$

The average battery lifespan is fewer than 500 hours



EXERCISE

Right-Tailed Test



$$\alpha = 0.05$$

$$z_{0.05} = 1.645$$

Null Hypothesis

$$H_0: \mu_1 \leq 500$$

The average battery lifespan is 500 hours at most

Alternative Hypothesis

$$H_a: \mu_1 > 500$$

The average battery lifespan is longer than 500 hours



EXERCISE

A manufacturing process is claimed to have an average defect rate of 10.32 units, with a known standard deviation of 3.17 units. The Statistical Process Control (SPC) department suspects this claim may no longer be valid and collects a random sample of 30 production units to test whether the true average defect rate differs significantly from 10.32.

Solution

dataset

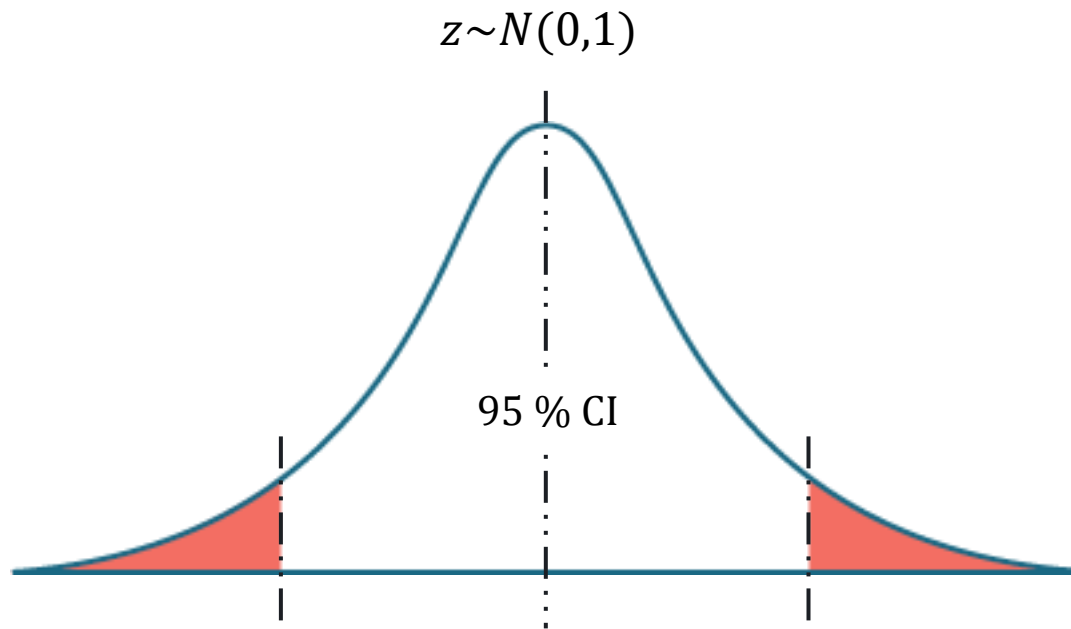
defects-30-sample.csv



EXERCISE

Two-Tailed Test

Solution



EXERCISE

A manufacturing process is claimed to have an average defect rate of 10.32 units, with a known standard deviation of 3.17 units. The Statistical Process Control (SPC) department suspects this claim may no longer be valid and collects a random sample of 30 production units to test whether the true average defect rate increases significantly from 10.32.

Solution

dataset

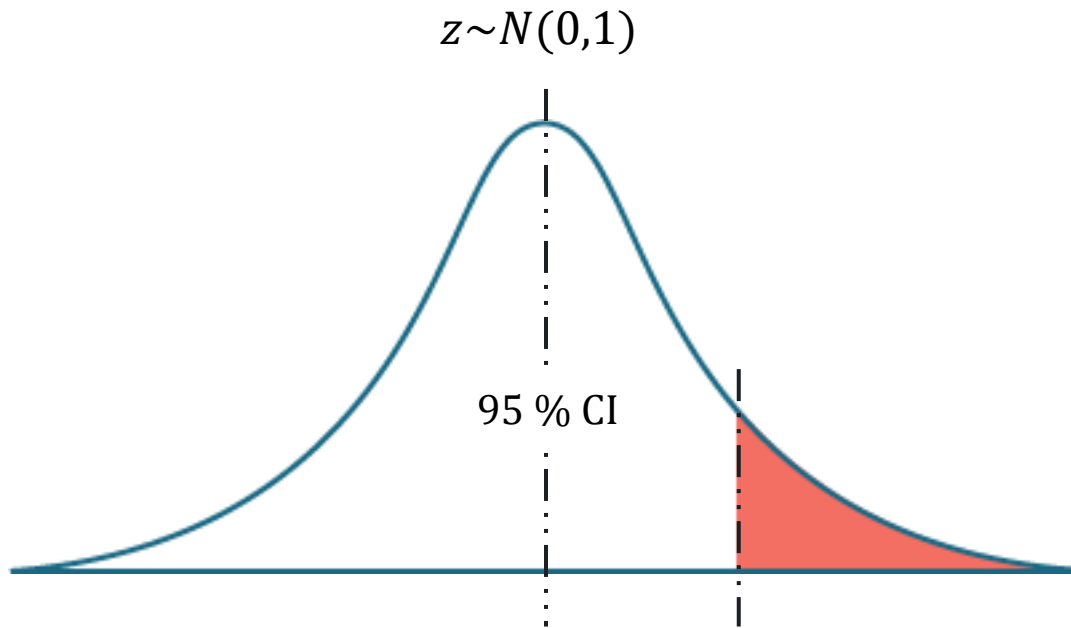
defects-30-sample.csv



EXERCISE

Right-Tailed Test

Solution



P-VALUE



P-VALUE

The **p-value** (probability value) is the **smallest level of significance** at which we can still reject the null hypothesis, given the observed sample statistic.

One-Tailed Test

p-value = 1 - value from the table

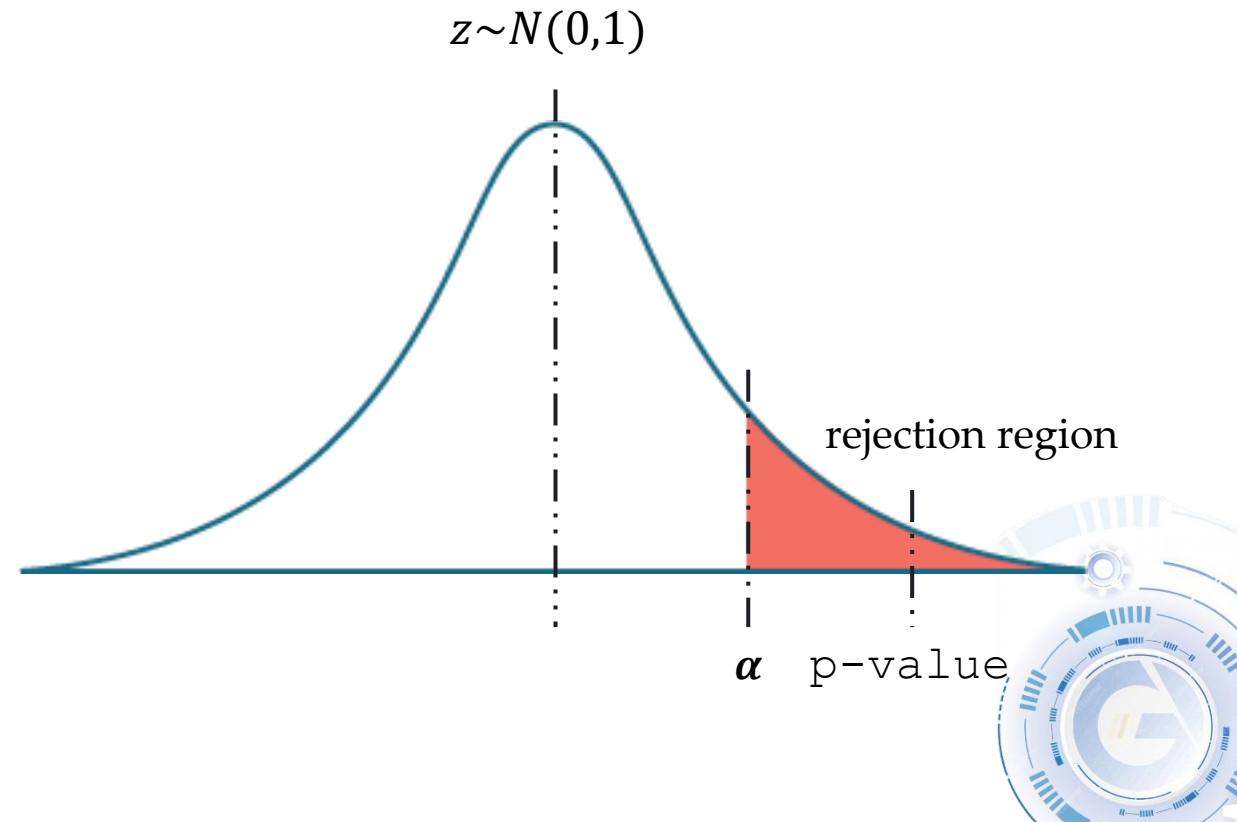
Two-Tailed Test

p-value = 2 (1 - value from the table)

Hypothesis Testing

Reject H_0 if **p-value** < α

Fail to reject H_0 if p-value $\geq \alpha$



P-VALUE

The p-value (probability value) is the smallest level of significance at which we can still reject the null hypothesis, given the observed sample statistic.

One-Tailed Test

p-value = 1 - value from the table

Two-Tailed Test

p-value = 2 (1 - value from the table)

syntax

```
from scipy import stats
```

One-Tailed Test

```
p_value = 1-stats.norm.cdf(z_score)
```

Two-Tailed Test

```
p_value = 2*(1-stats.norm.cdf(z_score))
```



EXERCISE

A manufacturing process is claimed to have an average defect rate of 10.32 units, with a known standard deviation of 3.17 units. The Statistical Process Control (SPC) department suspects this claim may no longer be valid and collects a random sample of 30 production units to test whether the true average defect rate differs significantly from 10.32.

Solution

dataset

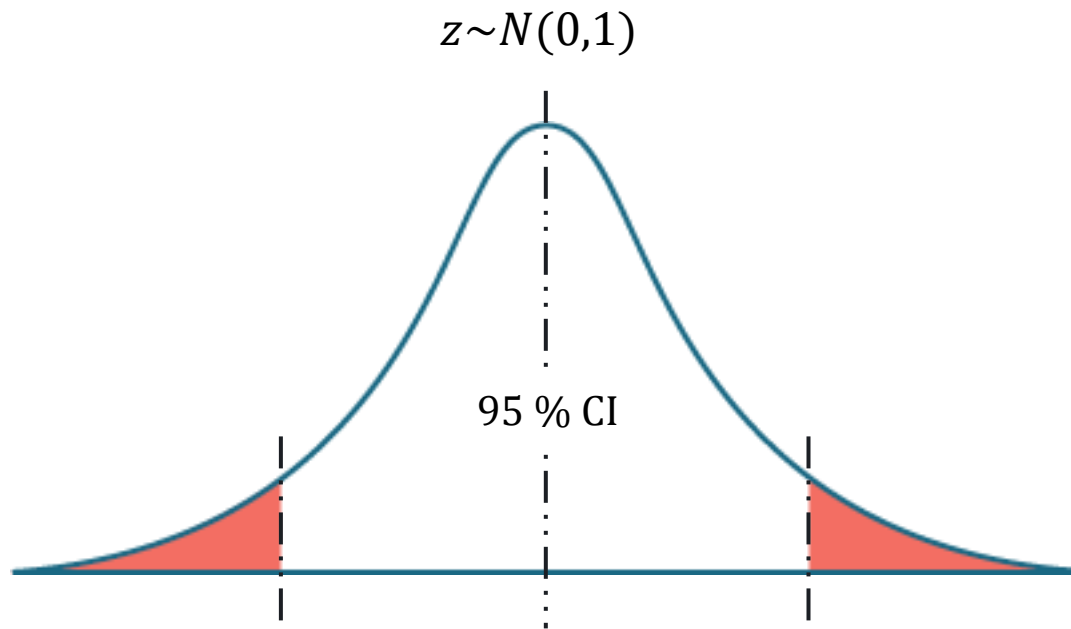
defects-30-sample.csv



EXERCISE

Two-Tailed Test

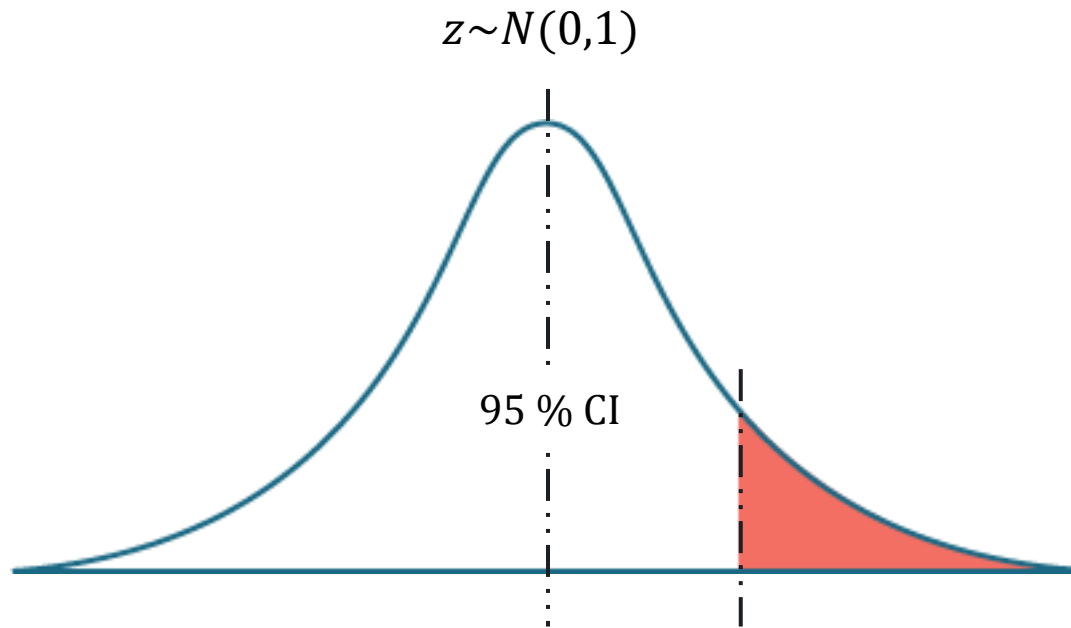
Solution



EXERCISE

Right-Tailed Test

Solution



LABORATORY

